Class 5:

1. Find the Fourier sine transform of $f(t) = \begin{cases} cos^2 t, & 0 < t < \frac{\pi}{2} \\ 0, & t > \frac{\pi}{2} \end{cases}$

Ans: $\frac{1}{2} cos\left(\frac{\pi\omega}{2}\right)\left[\frac{\omega}{(\omega^2-4)} - \frac{1}{\omega}\right] + \frac{1}{2\omega}$

2. Find the Fourier sine transformation of $f(t) = t^{m-1}$ where $m \neq 0$ or negative integer.

Hence find $F_s\left(\frac{1}{\sqrt{t}}\right)$.

Ans: $F_s(t^{m-1}) = \frac{\Gamma(m)}{s^m}\sin\left(\frac{m\pi}{2}\right); F_s\left(\frac{1}{\sqrt{t}}\right) = \sqrt{\frac{\pi}{2s}}$

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